

Detailed Project Document

Request for Amazon Web Services resources through GRNET

Project name	US-Foundation: Scaling Self-Supervised Ultrasound Foundation Models for Generalizable Multi-Task Biometry
Research field	Biomedical Engineering – Medical Image Analysis and Artificial Intelligence
Scientific supervisor	Prof. George K. Matsopoulos, Professor, School of Electrical and Computer Engineering, National Technical University of Athens (gmatsopoulos@biomed.ntua.gr)
Institution / laboratory	Biomedical Engineering Laboratory, School of Electrical and Computer Engineering, National Technical University of Athens (NTUA), Athens, Greece
Research team	Alexandros Barmperis (PhD candidate, technical lead; NTUA and Archimedes Research Unit, Athena Research Center; amparmperis@biomed.ntua.gr), Anna Panagiotakopoulou (NTUA), Vasileios E. Katsigiannis (NTUA)
Requested AWS services	Amazon EC2: one <code>p4de.24xlarge</code> (8× NVIDIA A100-SXM4-80GB) On-Demand, a <code>t3.large</code> control node and a 12-hour <code>p5.4xlarge</code> (H100) pilot; Amazon EBS (gp3); Amazon S3 (Standard); internet data transfer
AWS Region	US East (N. Virginia), <code>us-east-1</code> – the lowest A100-80GB price of all AWS Regions; fallback US West (Oregon), <code>us-west-2</code> , at the same price
Duration	6 months (renewable), counted from the activation of the credits
Estimated cost	\$17,580 = € 15,483 at the ECB reference rate of 2026-09-29 (1 € = 1.1355 US\$); requested amount: € 15,483
Cost estimate	[AWS Pricing Calculator link – create it from Appendix A and paste it here]

1. Research project description

Summary. Ultrasound is the most widely used imaging modality in pregnancy care and cardiology because it is safe, portable and inexpensive. Many clinical decisions rest on a few measurements taken on the image: the head and abdominal circumference and the femur length of a fetus, which track its growth; the angle of progression, which tracks labour; the size of the heart chambers and of the inferior vena cava. Each measurement is computed from a handful of anatomical *landmarks* that a sonographer places by hand, which is slow, varies between operators, and is sensitive to the scanner used (Salomon et al., 2019; Ghi et al., 2018; Mitchell et al., 2019). This project develops *one* artificial-intelligence model that places these landmarks automatically for nine different clinical tasks, and studies how to make it generalize across hospitals, scanners and tasks. The model starts from a general-purpose vision *foundation model* (DINOv2, trained by Meta on 142 million natural images; Quab et al., 2023), adapts it to ultrasound with 191,170 unlabeled frames through self-supervised learning, which needs no manual labels, and then learns to place landmarks from 6,768 labeled images. The first version was built for the MICCAI 2026 Foundation Model Challenge for Ultrasound Biometry (Bai et al., 2026a) and is described in our camera-ready challenge paper (Panagiotakopoulou et al., 2026). The requested AWS credits fund the controlled, repeated experiments that are needed to turn this prototype into a reliable, openly released ultrasound landmark foundation model.

Significance: impact and timeliness. *Clinically*, automated biometry shortens examinations and removes inter-observer variability, and a single model for prenatal, intrapartum and cardiac/vascular tasks is far simpler to validate and deploy than nine task-specific tools; multi-centre benchmarks show that site and scanner differences degrade single-task models (Di Vece et al., 2026; Bai et al., 2026b). *Scientifically*, ultrasound foundation models are only now emerging (Jiao et al., 2024; Ambsdorf et al., 2025; Lin et al., 2025), and they target classification, segmentation or few-shot analysis; unified

sparse landmark localization across biometric domains is largely unexplored, and the new nine-task benchmark (Bai et al., 2026a; Deng et al., 2026) makes it measurable for the first time. The work is *timely* because our challenge entry exposed two questions that decide whether such models can be trusted and that can only be settled by large, controlled experiments: how self-supervised adaptation should be scheduled, and why gains measured internally did not transfer to the hidden test set.

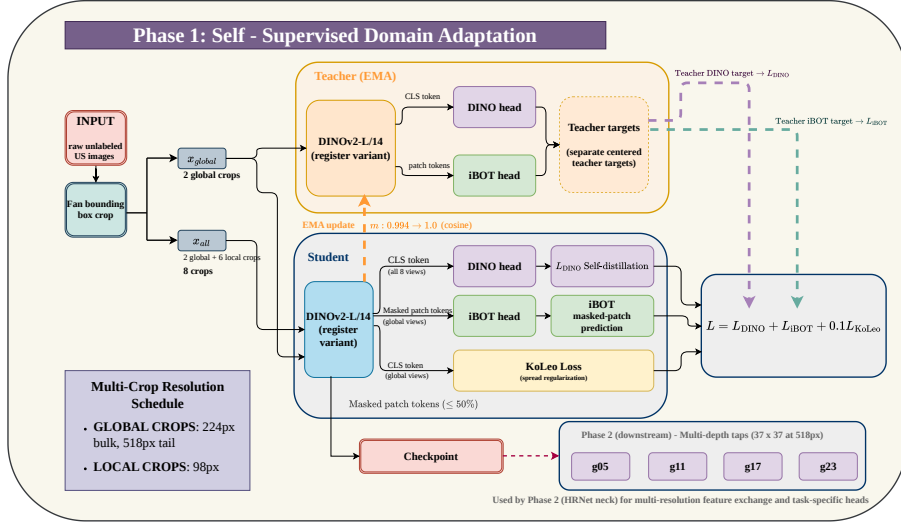
How the framework works (in brief). The pipeline (Fig. 1) separates domain-level representation learning from task-specific localization. *Phase 1* adapts the register variant of DINOv2-L/14 (Darcet et al., 2024) to unlabeled ultrasound by continued self-supervised pretraining that follows the DINOv2 recipe: CLS-token self-distillation (Caron et al., 2021), masked patch-token prediction (Zhou et al., 2022) and a KoLeo spread regularizer (Sablayrolles et al., 2019), with an exponential-moving-average (EMA) teacher, 2 global and 6 local crops, 65,536 prototypes and an effective batch of 256. Training runs mostly at a cheap 224-px resolution and ends with a short 518-px “tail” at the deployment resolution; the augmentations are ultrasound-specific (crops restricted to the imaging fan, gentle $\pm 10^\circ$ rotations, masking restricted to tissue). *Phase 2* feeds a 518×518 letterboxed image through the adapted encoder, taps four depths (blocks 5, 11, 17, 23), and fuses them in an HRNet-style multi-resolution neck (Sun et al., 2019) with branches at 37×37, 74×74 and 148×148 and GroupNorm throughout. Nine task-specific heads, selected by the task of each batch, output landmark heat-maps that a soft-argmax (Nibali et al., 2018) turns into sub-pixel coordinates, mapped back to original-image pixels. The model is trained with a masked L_1 loss in original-pixel units, a task-balanced sampler and layer-wise learning-rate decay, and checkpoints are selected on a criterion aligned with the official challenge metric: 50 % landmark error plus 50 % clinical-measurement error, macro-averaged with equal weight per task.

What is already established. On the official validation set (619 images, nine tasks) our single model reaches a mean radial error (MRE) of 26.34 px and a measurement error (MAE) of 29.70; the best leaderboard entry reported 22.56 px / 29.02. On the hidden test set the final submission obtained 30.66 px / 36.78, and one task degraded sharply: the angle-of-progression MRE rose from 16.0 px (validation) to 65.0 px (test). Frozen-encoder probes (Fig. 2) show that ultrasound self-supervision is useful but *non-monotonic*: 20 epochs at 224 px lower the MRE from 31.27 px (no adaptation) to 24.70 px, whereas 100 epochs score worse than no adaptation on the selection criterion; a 4-epoch 518-px tail recovers the best score of all checkpoints (0.0747 vs. 0.0974 without the tail). Full-resolution training is not better at matched epochs and costs $4.2\times$ more per epoch. Two further observations limit how far these conclusions can be trusted: (i) a five-fold comparison of decoders gave differences (0.0707 ± 0.0089 vs. 0.0740 ± 0.0068) smaller than the fold-to-fold spread, and (ii) our internal audit found that image-level folds leak cine loops, since in fold 0, 45 % of the A4C and 24 % of the PLAX validation frames come from a loop that also has frames in training. Leakage of this kind is a documented cause of over-optimistic medical-imaging results (Varoquaux and Cheplygina, 2022; Kapoor and Narayanan, 2023), and is a plausible reason why internal gains did not transfer to the hidden test.

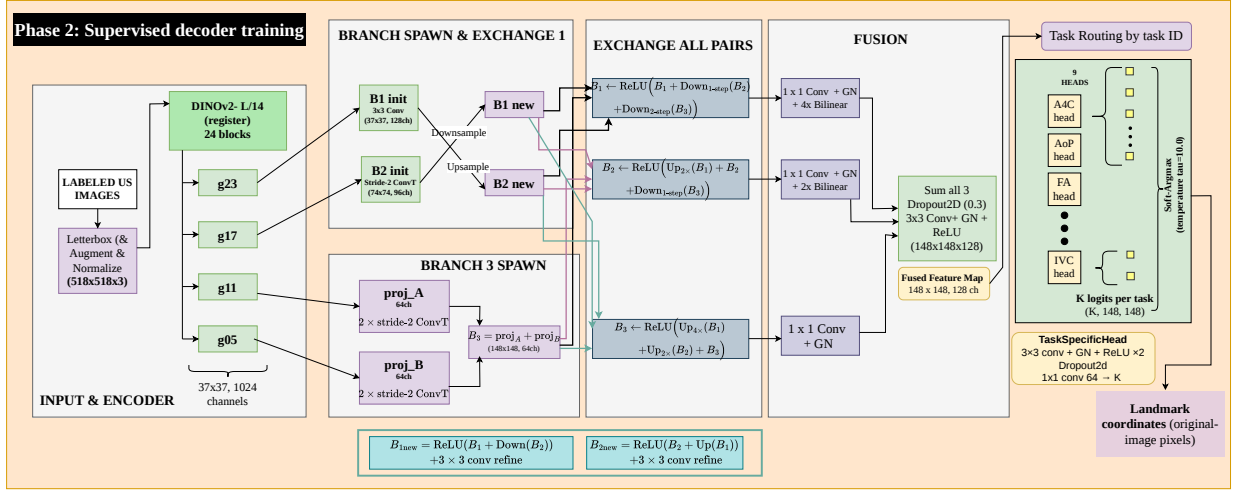
Scientific questions. The project is organized around four questions.

- (Q1) *Robust evaluation.* Which of the conclusions above survive a leakage-free, cine-loop-grouped five-fold cross-validation, and does the leakage-free internal score predict the hidden-test ranking?
- (Q2) *Self-supervised schedule and data.* How long, at which resolution, and on how much unlabeled data should a general-purpose encoder be adapted to ultrasound? Is the non-monotonic duration curve reproducible over seeds, and where should the high-resolution tail start and how long should it last?
- (Q3) *Capacity.* Does the benefit of in-domain self-supervision grow or shrink with encoder size, from ViT-B/14 (87 M parameters) through ViT-L/14 (304 M) to ViT-g/14 (1.14 B) (Dosovitskiy et al., 2021; Zhai et al., 2022)?
- (Q4) *Phase-2 design.* Which decoder and training choices (multi-level fusion, fine-tuning depth from 0 to 24 blocks, measurement-aware and robust losses, task-sampling temperature) change the official metric by more than the fold-and-seed noise?

The deliverable is an openly released ultrasound landmark foundation model (five-fold ensemble with



(a) Phase 1: self-supervised domain adaptation of DINOv2-L/14 on 191,170 unlabeled frames.



(b) Phase 2: multi-depth taps, HRNet-style neck, task routing to nine heads and soft-argmax decoding.

Figure 1: The two-phase pipeline (figures reproduced from Panagiotakopoulou et al., 2026).

test-time augmentation and ensemble-based uncertainty; Lakshminarayanan et al., 2017), its code, and a leakage-free evaluation protocol for the nine-task benchmark.

Why numerical experiments are the effective tool. None of these questions can be answered analytically: the behaviour of a 304 M–1.14 B-parameter transformer adapted on 191,170 images can only be measured, by training many variants under identical conditions and repeating them over folds and seeds, because the fold-to-fold spread (13–15 %) is as large as the effects of interest (Bouthillier et al., 2021). One self-supervised run of the published recipe took 95.7 A100-hours, a single fine-tuning run takes 8.8 A100-hours on average, and the full plan comprises 396 GPU jobs (2,920 A100-hours at measured rates). This is far beyond what our shared hardware can deliver in six months (Section 4).

2. Work plan and management of the requested resources

Strategy. All jobs are independent single-GPU runs (folds, seeds and configurations), so the campaign maps directly onto one `p4de.24xlarge` node with eight A100-80GB GPUs, the same GPU model on which every timing in this document was measured. The node is run in *bursts*: it is started only when a queue of at least eight jobs is ready, runs eight jobs concurrently, and is stopped as soon as the queue drains, so that no GPU-hour is paid while idle. A small `t3.large` control node holds the job queue, synchronizes results and monitors spending. Datasets, checkpoints and logs live in

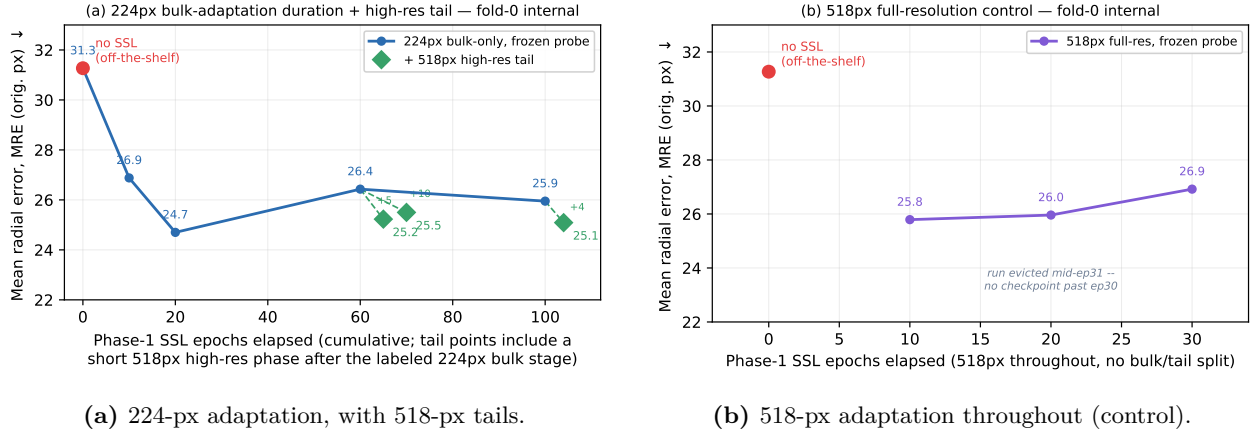


Figure 2: Frozen-encoder probe MRE (original px) versus self-supervised adaptation length, fold 0, single seed (Panagiotakopoulou et al., 2026). The non-monotonic curve is the central finding that WP2 must confirm over seeds and leakage-free folds.

Amazon S3, and each run writes an atomic `latest_checkpoint` every epoch, so any interruption costs at most one epoch. The longest single job, the ViT-g self-supervised run (216 budgeted hours, 9.0 days), fixes the length of the largest burst; the other seven GPUs are filled with WP2–WP4 jobs during it.

Work packages. Budgeted A100-hours include the 25 % per-job slack of Section 6.

- **WP0 – Set-up, timing parity, H100 pilot (M1, 19 A100-h).** Account, quota (96 vCPUs of On-Demand P instances), S3 and EBS set-up; upload of the 20 GB dataset and existing checkpoints; one SSL epoch at each resolution, one fine-tuning run and one probe, to confirm that AWS timings match Table 3 within $\pm 10\%$; a 12-hour benchmark on one H100 (p5.4xlarge) to decide whether single-GPU H100 instances would be cheaper for the longest jobs.
- **WP1 – Leakage-free re-baselining, Q1 (M1–M2, 501 A100-h).** New cine-loop-grouped five-fold splits; six reference Phase-2 configurations $\times$ 5 folds; frozen probes of the 12 existing encoders $\times$ 5 folds; out-of-fold predictions for every ensemble, compared against our hidden-test results.
- **WP2 – SSL schedule, resolution and data, Q2 (M2–M4, 1,012 A100-h).** Two new seeds of the 224-px trajectory with 518-px tails from epochs 20 and 60; a sweep of tail start (epochs 20/40/60) and length (2/5/10 epochs); continuation of the full-resolution control, which was evicted at epoch 30 on our shared node, to epoch 60; unlabeled-data scaling with 10/25/50 % subsets at matched compute; five-fold probes of all 22 new checkpoints and five-fold fine-tuning of the three best.
- **WP3 – Encoder capacity, Q3 (M3–M5, 865 A100-h).** The same protocol for ViT-B/14 and ViT-g/14 (60 bulk epochs, tails from epochs 20 and 60, five-fold probes, five-fold fine-tuning with and without adaptation); ViT-L results come from WP1–WP2.
- **WP4 – Phase-2 design and objectives, Q4 (M4–M5, 895 A100-h).** Fine-tuning depth of 8, 12 and 24 blocks; two alternative tap sets; measurement-aware, Wing and smooth- L_1 losses; sampler temperature and augmentation strength; two extra seeds of the best configuration, all over five leakage-free folds.
- **WP5 – Final ensembles, uncertainty and release (M5–M6, 357 A100-h).** Final ViT-L ensemble (5 folds $\times$ 3 seeds) and ViT-g ensemble (5 folds), all-data models, multi-scale/intensity test-time augmentation, ensemble-spread uncertainty, Docker image, public code and weights, final report.

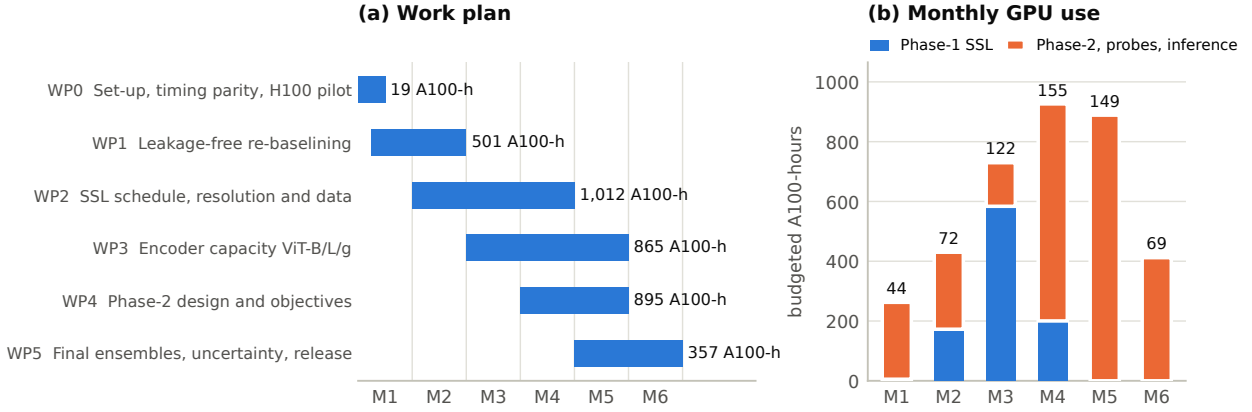


Figure 3: (a) Six-month schedule with the budgeted A100-hours of each work package. (b) Monthly budgeted A100-hours split into self-supervised pretraining (Phase 1) and supervised runs, probes and inference (Phase 2); the label on each column is the `p4de.24xlarge` node-hours of that month, including packing and contingency.

Table 1: Monthly resource consumption. A100-hours are budgeted (with slack); node-hours and cost include packing and contingency, and the cost includes the storage, control-node and pilot lines of Table 6.

Month	Main activities	SSL A100-h	Phase-2 A100-h	Node-h	USD	EUR
M1	WP0 set-up and parity; WP1 starts	5	257	44	\$1,408	€1,240
M2	WP1 completes; WP2 SSL runs start	173	258	72	\$2,097	€1,846
M3	WP2 SSL + probes; WP3 SSL (ViT-B, ViT-g)	584	146	122	\$3,475	€3,060
M4	WP2 Phase-2; WP3 probes/Phase-2; WP4 starts	200	726	155	\$4,381	€3,858
M5	WP3 and WP4 complete; WP5 starts	0	889	149	\$4,207	€3,705
M6	WP5 final ensembles, inference, release, report	0	412	69	\$2,013	€1,773
Total		962	2,687	612	\$17,580	€15,483

3. Numerical methods, algorithms, codes and libraries

Numerical methods and algorithms. The numerical core is stochastic-gradient training of vision transformers (Dosovitskiy et al., 2021) with mixed-precision (bfloat16) arithmetic, AdamW, cosine learning-rate, weight-decay and momentum schedules, gradient accumulation to an effective batch of 256, gradient clipping and EMA teachers. Phase 1 implements self-distillation with centred and sharpened teacher targets over 65,536 prototypes (Caron et al., 2021), block-wise masked patch prediction (Zhou et al., 2022) and the Kozachenko–Leonenko entropy estimator (Sablayrolles et al., 2019), as in DINOv2 (Oquab et al., 2023). Phase 2 implements multi-resolution feature exchange (Sun et al., 2019), differentiable soft-argmax coordinate regression with an optional DSNT regularizer (Nibali et al., 2018), masked L_1 /Wing/smooth- L_1 losses re-weighted to original-image pixels, layer-wise learning-rate decay, and a task-homogeneous batch sampler with temperature-controlled task frequencies (every forward pass runs exactly one task head). Inference inverts the letterbox transform exactly and averages predictions over ensemble members and multi-scale/intensity views, with mean, median or hierarchical (per-member, then across-member) reduction.

Evaluation protocol and statistics. Every comparison uses five cine-loop-grouped folds; headline comparisons use three seeds (Bouthillier et al., 2021). We report the official metric, landmark MRE in original pixels plus clinical-measurement error (angle in degrees for AoP, lengths in pixels elsewhere), macro-averaged over the nine tasks. Our scorer reproduces the official evaluation outputs

exactly, and checkpoints are selected on this metric, never on the training loss. Differences are tested with paired bootstrap confidence intervals and Wilcoxon signed-rank tests over images, with Holm correction. Encoder quality is measured separately from fine-tuning with frozen-encoder probes, because self-supervised loss values are not comparable across recipes.

Codes, packages and libraries. All experiments run through our Python package `gubiometry` (single CLI: `make_splits`, `phase1`, `phase2`, `kfold`, `predict`, `evaluate`), driven by one YAML file per run with command-line overrides; every run stores its full configuration and the MD5 hash of its Phase-1 checkpoint, so any model can be rebuilt exactly. Dependencies: PyTorch 2.x with CUDA 12 and cuDNN, DINOv2 through `torch.hub` (Apache-2.0), NumPy, OpenCV, Albumentations, PyYAML, TensorBoard and matplotlib. A Docker image of the inference pipeline already exists; it produced our final hidden-test submission fully offline. Code: <https://github.com/abarmper/US-Foundation> [confirm the repository is public before sending].

How the methods enable the research. One code path trains every configuration identically, so each difference can be attributed to the single factor varied (Q2–Q4). Loop-grouped folds and seeds make the differences statistically testable (Q1). Because encoder size, resolution, tail length, data fraction and fine-tuning depth are all configuration parameters, the whole plan is a grid of independent single-GPU jobs, which is exactly the workload an eight-GPU cloud node runs efficiently.

4. Why AWS – suitability and scaling evidence

Current systems and their limits. All results so far were produced on an $8\times$ A100-SXM4-80GB cloud node provided by the NVIDIA Academic Hardware Grant Program (MedViLA) and shared by several researchers. In practice we could use one or two GPUs at a time: when this document was written, all eight GPUs were occupied by jobs of other projects (3D segmentation and image-synthesis studies of the laboratory and its collaborators). Long jobs are fragile on a shared node. The 518-px control run had to be restarted six times after external interruptions and was abandoned at epoch 30 of 100, which is why that comparison in our paper stops early. The grant is also time-limited [confirm the end date of the NVIDIA grant allocation]. At the budgeted 3,649 A100-hours, the plan would occupy two dedicated GPUs for 76 days, and more than that on a contended node. This is incompatible with a six-month timeline.

Why AWS, and which instance. The `p4de.24xlarge` provides the *same* GPU (A100-SXM4-80GB) on which all our timings were measured, so the unit costs of Table 3 transfer without extrapolation. It also has 96 vCPUs, 1,152 GiB of memory and 8×1 TB local NVMe, enough for eight concurrent data-loading pipelines. The 80 GB memory is required: our measurements (Table 2) show 60 GB for ViT-L self-supervision at 518 px, and ViT-g needs 58–62 GB even with micro-batches reduced to 16 (224 px) and 4 (518 px); micro-batches of 32 and 8 do not fit in 80 GB at all. The cheaper A100-40GB node (`p4d`, \$2.74 per GPU-hour) therefore cannot run the 518-px tail or any ViT-g job at our settings. A single-GPU H100 (`p5.4xlarge`, \$6.88/h) costs $2.0\times$ an A100 GPU-hour, so it only pays off for jobs that run at least $2.0\times$ faster; the WP0 pilot measures this instead of assuming it.

Which Region. Figure 4 compares the On-Demand price per GPU-hour of both A100 instances in every Region that offers them (AWS price list of 2026-09-25). The A100-80GB node is cheapest in `us-east-1` and `us-west-2` (\$27.44705 per hour, i.e. \$3.43 per GPU-hour). We choose `us-east-1` and keep `us-west-2` as a same-price fallback. The challenge data are de-identified research data and no personal data are processed. Should an EU Region be preferred, `eu-central-1` (Frankfurt) offers the same node at \$34.29/h (+24.9%), which would add about \$4,189 to the estimate.

Scaling evidence. Table 2 and Fig. 5 report single-GPU throughput for the three encoder sizes, measured with our own training code on an idle A100-80GB. For ViT-L the benchmark agrees with the production logs within 4% for self-supervision; the fine-tuning log value is lower only because it includes the per-epoch validation pass. Cost grows *sub-linearly* with model size: ViT-g has $3.7\times$ the parameters of ViT-L but costs $3.5\times$ for self-supervision and $2.5\times$ for fine-tuning, because the frozen part of the encoder only runs forward. ViT-B costs $0.38\times$ and $0.56\times$ of ViT-L. Resolution is the other lever: a 518-px global crop has $5.3\times$ the patch tokens of a 224-px crop, but an epoch

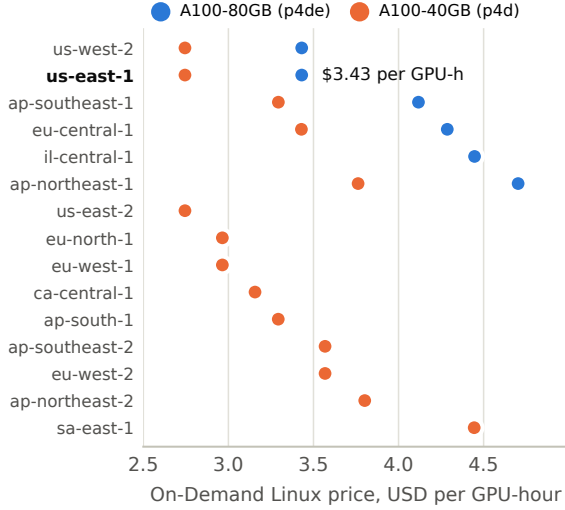


Figure 4: On-Demand Linux price per GPU-hour of the A100-80GB (p4de.24xlarge) and A100-40GB (p4d.24xlarge) instances in every AWS Region that offers them, from the public AWS price list of 2026-09-25. **us-east-1** (bold) and **us-west-2** share the lowest A100-80GB price, \$3.43 per GPU-hour; the A100-80GB instance is offered in six Regions only.

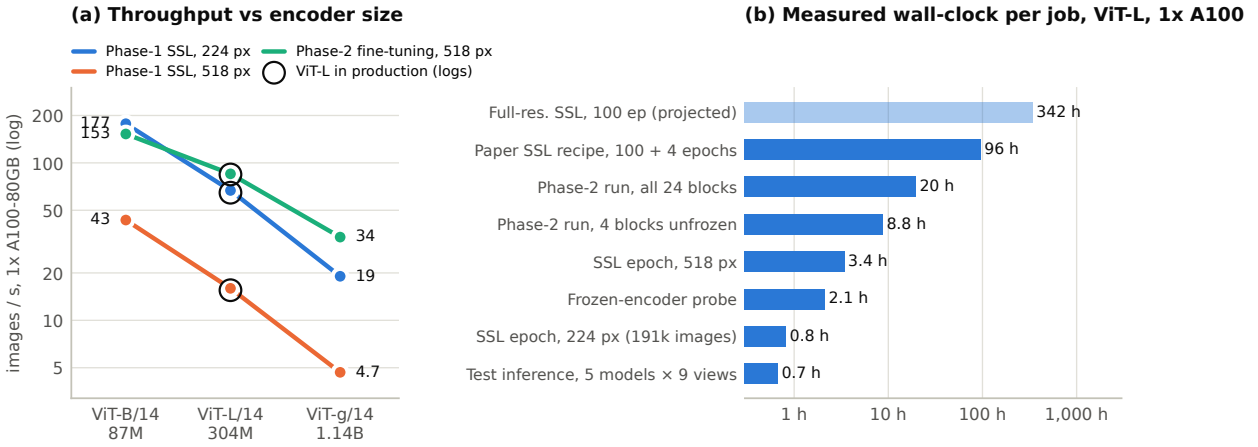


Figure 5: Measured scaling on one A100-80GB. (a) Throughput versus encoder size for self-supervised pretraining (224 and 518 px) and Phase-2 fine-tuning (log-log); black rings mark the ViT-L throughput of our production runs. (b) Measured wall-clock of each job type for ViT-L (log scale; the full-resolution run is projected from its measured epochs because it was never allowed to finish).

costs only 4.2× more, and our curriculum spends 96 % of its epochs at the cheap resolution. Across jobs the scaling is linear: the jobs are independent, so eight GPUs complete the budgeted plan in 612 node-hours (25.5 days of node time), against 152 days on one GPU.

Table 2: Throughput in images/s on one A100-80GB (peak memory in brackets), same code path, bfloat16. Micro-batch 32 at 224px and 16 at 518px for SSL (“mb” where ViT-g needed less), 64 for Phase 2. Production values are the ViT-L means from our training logs (Phase-2 values include validation). The last column is the time cost of ViT-g relative to ViT-L.

Workload	ViT-B/14 (87M)	ViT-L/14 (304M)	ViT-L, logs	ViT-g/14 (1.14B)	ViT-g cost
SSL, 224 px	177.3 (16 GB)	66.8 (36 GB)	64.6	19.0 (62 GB, mb 16)	3.51×
SSL, 518 px	43.3 (27 GB)	16.0 (60 GB)	15.5	4.7 (58 GB, mb 4)	3.41×
Fine-tuning	152.6 (24 GB)	85.2 (29 GB)	78.9	33.8 (44 GB)	2.52×
Frozen probe	208.1 (13 GB)	109.0 (14 GB)	97.8	40.6 (18 GB)	2.69×

How the AWS credits will help.

- *They buy exactly the compute the plan needs:* 612 node-hours of $8 \times$ A100-80GB, i.e. 4,896 A100-hours, at \$4.60 per budgeted A100-hour including packing and contingency. That is \$50 per fine-tuning run, \$12 per probe and \$551 per full self-supervised recipe, so every item in Table 4 has a known price.
- *Dedicated GPUs, no evictions:* eight GPUs per burst shorten the programme from months of shared-node time to 25.5 days of node time, and uninterrupted multi-day jobs (the 9.0-day ViT-g run) become possible for the first time.
- *New science:* the capacity study of WP3 (ViT-B and ViT-g, \$3,983) and the leakage-free, multi-seed protocol of WP1–WP2 (\$2,307 + \$4,657) cannot be run on our current allocation.
- *Reproducibility and accountability:* S3 keeps every checkpoint, configuration and log; each resource is tagged with its work package, so the spend per work package can be reported to GRNET.

5. Experience using cloud and HPC resources

The team trains and serves all its models on GPU cloud instances. On the current $8 \times$ A100 NVIDIA cloud node we manage our own environments, multi-GPU job chains, checkpoint/resume, and the Codabench Docker image that ran our final submission fully offline. We have also provisioned rented GPU instances (RunPod), including building the CUDA/PyTorch toolchain, and our laboratory uses a Slurm-managed GPU cluster for other projects. We have been allocated and operated compute by GRNET before, through its educational AWS programme, on which we provisioned and ran GPU compute on EC2 instances, so we are familiar with GRNET’s allocation process and with EC2, EBS and S3. The workload needs no MPI and no inter-node communication: jobs are arrays of single-GPU runs, launched from YAML configurations and made robust by atomic per-epoch checkpoints with automatic resume, a scheme we already rely on.

6. Justification of the requested budget

From measured timings to credits. Every unit cost in Table 3 is measured from our A100-SXM4-80GB training logs (for example, 100 epochs of 224-px self-supervision took 82.1 h), and every job of Table 4 is expressed in those units; ViT-B and ViT-g jobs are scaled by the measured ratios of Table 2. The work plan totals 2,920 A100-hours at measured rates. Slack is then added in three explicit steps (Table 5). (1) 25 % on *every* job covers differences in CPU and data loading on AWS (12 vCPUs per GPU instead of 32 on our node), early-stopping variability (runs last 45–150 epochs; we budget the mean), and checkpoint I/O. (2) Division by 90 % packing efficiency pays for GPU slots that stay idle while the node is up, since `p4de` bills all eight GPUs. (3) A 20 % contingency covers failed or repeated runs, debugging, extra seeds where a comparison turns out to be borderline, and changes in AWS prices or in the exchange rate during the allocation. The result, 612 node-hours, is rounded up to whole hours per month so that the AWS Pricing Calculator reproduces it exactly.

Table 3: Unit costs on one A100-80GB: measured from our training logs, then budgeted with +25 % slack; the cost column uses the effective \$4.60 per budgeted A100-hour (packing and contingency included).

Job (ViT-L unless stated)	Measured h	Budgeted h	Cost	Source
SSL epoch, 224 px bulk (191,170 images)	0.82	1.03	\$5	100 epochs, phase1_dinov2
SSL epoch, 518 px (tail / full-resolution)	3.42	4.27	\$20	45 epochs, 3 runs
Paper SSL recipe (100 ep @224 + 4 ep @518)	95.7	119.6	\$551	start to finish
Phase-2 run, 4 blocks unfrozen	8.76	10.94	\$50	mean of 22 runs, 45–150 ep
Phase-2 run, all 24 blocks unfrozen	19.59	24.49	\$113	mean of 5 runs
Frozen-encoder probe, ≤ 25 ep	2.12	2.65	\$12	mean of 14 runs
OOF inference, 6,743 images, 9 TTA views	1.86	2.32	\$11	0.99 s/image measured
Test inference, 5 models $\times$ 9 views, 619 images	0.67	0.84	\$4	measured

Table 4: Work plan in A100-hours: number of jobs, measured hours per job, total at measured rates, and budgeted total with +25 % slack. ViT-B/ViT-g jobs are scaled by the measured ratios of Table 2.

Job	#	h/job	Measured	Budgeted
WP0 – Set-up, timing parity, H100 pilot				
Parity: 1 SSL epoch at 224 px + 1 at 518 px (ViT-L)	1	4.2	4	5
Parity: 1 Phase-2 run + 1 frozen probe (fold 0)	1	10.9	11	14
<i>WP0 subtotal</i>			15	19
WP1 – Leakage-free re-baselining (Q1)				
Phase-2, 6 reference configurations $\times$ 5 loop-grouped folds	30	8.8	263	328
Frozen probes, 12 existing encoders $\times$ 5 folds	60	2.1	127	159
Out-of-fold inference, 9-view TTA, 6 ensembles	6	1.9	11	14
<i>WP1 subtotal</i>			401	501
WP2 – SSL schedule, resolution and data (Q2)				
SSL bulk, 60 ep at 224 px, 2 new seeds	2	49.2	98	123
518-px tails (5 ep) from ep20 and ep60, 2 new seeds	4	17.1	68	85
Tail sweep, seed 42: ep20/ep40 + 5 ep; ep20 + 2/10 ep	4	18.8	75	94
Full-resolution control, 518 px, ep31 $\rightarrow$ ep60	1	102.5	102	128
Unlabeled-data scaling: 10/25/50 % subsets (20 + 5 ep)	3	33.5	100	126
Frozen probes, 22 new checkpoints $\times$ 5 folds	110	2.1	233	292
Phase-2, top-3 SSL checkpoints $\times$ 5 folds	15	8.8	131	164
<i>WP2 subtotal</i>			809	1,012
WP3 – Encoder capacity ViT-B/L/g (Q3)				
ViT-B: SSL bulk, 60 ep at 224 px	1	18.6	19	23
ViT-B: 518-px tails (5 ep) from ep20 and ep60	2	6.3	13	16
ViT-B: frozen probes, 3 encoders $\times$ 5 folds	15	1.1	17	21
ViT-B: Phase-2, no-SSL vs SSL $\times$ 5 folds	10	4.9	49	61
ViT-g: SSL bulk, 60 ep at 224 px	1	172.7	173	216
ViT-g: 518-px tails (5 ep) from ep20 and ep60	2	58.3	117	146
ViT-g: frozen probes, 3 encoders $\times$ 5 folds	15	5.7	85	107
ViT-g: Phase-2, no-SSL vs SSL $\times$ 5 folds	10	22.1	221	276
<i>WP3 subtotal</i>			692	865
WP4 – Phase-2 design and objectives (Q4)				
Fine-tuning depth: 8 blocks $\times$ 5 folds	5	10.8	54	67
Fine-tuning depth: 12 blocks $\times$ 5 folds	5	12.8	64	80
Fine-tuning depth: all 24 blocks $\times$ 5 folds	5	19.6	98	122
Multi-level taps (2 variants) $\times$ 5 folds	10	8.8	88	109
Losses: measurement-aware (2 weights), Wing, smooth-L1 $\times$ 5 folds	20	8.8	175	219
Sampler temperature (0, 1) + strong augmentation $\times$ 5 folds	15	8.8	131	164
Best configuration, 2 extra seeds $\times$ 5 folds	10	8.8	88	109
Out-of-fold inference, 10 ensembles	10	1.9	19	23
<i>WP4 subtotal</i>			716	895
WP5 – Final ensembles, uncertainty, release				
ViT-L final: 5 folds $\times$ 3 seeds + all-data model	16	8.8	140	175
ViT-g final: 5 folds + all-data model	6	22.1	132	166
OOF + test inference, multi-scale/intensity TTA (4 ensembles)	1	13.2	13	17
<i>WP5 subtotal</i>			286	357
Total	396		2,920	3,649

Table 5: From measured A100-hours to billed p4de.24xlarge node-hours.

Step	A100-hours	Node-hours
Measured-rate A100-hours of the work plan	2,920	–
+25 % slack on every job	3,649	–
÷ 0.90 packing (8 GPUs billed per node)	4,055	507
+20 % contingency, rounded up to whole hours/month	4,896	612

Table 6: Estimated cost by AWS service in us-east-1 for the six-month allocation (AWS price list of 2026-09-25; On-Demand, taxes excluded). Euro amounts use the ECB reference rate of 2026-09-29, 1 € = 1.1355 US\$. These lines are the entries of the AWS Pricing Calculator estimate (Appendix A).

Service	Configuration	Quantity	Unit price	USD	EUR
Amazon EC2	p4de.24xlarge, On-Demand Linux (8x A100-80GB)	612 instance-h	\$27.44705	\$16,797.59	€14,793.13
Amazon EC2	p5.4xlarge, On-Demand Linux (1x H100-80GB), WP0 pilot	12 instance-h	\$6.88	\$82.56	€72.71
Amazon EC2	t3.large, On-Demand Linux (control/staging node)	600 instance-h	\$0.0832	\$49.92	€43.96
Amazon EBS	gp3, 1,000 GB (3,000 IOPS, 125 MB/s baseline)	6,000 GB-month	\$0.08	\$480.00	€422.72
Amazon S3	Standard, 1,000 GB average (data, checkpoints, logs, container image)	6,000 GB-month	\$0.023	\$138.00	€121.53
Amazon S3	Requests: 100k PUT/LIST + 1M GET per month	6 month	\$0.9	\$5.40	€4.76
Data transfer	Internet egress (models, predictions, logs)	300 GB	\$0.09	\$27.00	€23.78
Total, 6 months (\$2,930.08 per month)				\$17,580.47	€15,482.58

Requested amount. The estimate is \$17,580, i.e. €15,483 at 1.1355 US\$/€, and we request exactly this amount, **€15,483**, for six months (\$2,930 or €2,580 per month on average). The GPU node accounts for 96 % of it, and its hours are fully itemized in Table 4. Storage is deliberately modest: a 1 TB gp3 volume for the working set, the node’s local NVMe as scratch, and S3 for about 1 TB of datasets and retained checkpoints. We keep only the selected checkpoint of each finished run (1.2 GB for ViT-L).

Minimum viable subset. If only part of the amount can be granted, WP0–WP2 plus the ViT-L part of WP5 (288 node-hours, \$8,688 $\approx$ €7,651, 49 % of the request) still answer Q1 and Q2 and deliver the released ViT-L model; the capacity study (Q3) and the Phase-2 design study (Q4) would then be dropped.

Cost control. Spending is monitored daily in AWS Cost Explorer, and AWS Budgets alerts fire at 50 %, 80 % and 100 % of each month’s plan (Table 1). The node runs only while its queue is full and is stopped by the control node when it drains. S3 lifecycle rules delete intermediate checkpoints after evaluation. Where Spot capacity of the same instance is available at a lower price, short probe and inference jobs use it; any saving extends the contingency, not the scope.

7. Data management, dissemination and acknowledgement

The challenge data are de-identified images released for research; they are kept in a private, encrypted S3 bucket (server-side encryption, TLS in transit) accessible only to the project’s IAM users, are never redistributed, and are deleted from AWS at the end of the allocation. Code, trained weights (within the Apache-2.0 terms of DINOv2) and the leakage-free fold definitions will be released openly with the resulting journal publication. As required by the programme, all publications that use these resources will state that “*the resources were granted with the support of GRNET*”, and their citation records will be sent to helpdesk@aws.grnet.gr.

References

- Jakob Ambsdorf, Asbjørn Munk, Sebastian Llabias, Anders N. Christensen, Kamil Mikolaj, Randall Balestriero, Martin G. Tolsgaard, Aasa Feragen, and Mads Nielsen. General methods make great domain-specific foundation models: A case-study on fetal ultrasound. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, volume 15966 of *Lecture Notes in Computer Science*, pages 271–281. Springer Nature Switzerland, 2025.
- J. Bai, M. Yaqub, J. Ma, K. Lekadir, J. Gan, W. Cai, D. Ni, and S. Li. Foundation model challenge for ultrasound biometry. Zenodo, 2026a. MICCAI 2026 challenge.
- J. Bai et al. Beyond benchmarks of IUGC: Rethinking requirements of deep learning method for intrapartum ultrasound biometry from fetal ultrasound videos. *Medical Image Analysis*, 2026b.
- Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Nazanin Mohammadi Sepahvand, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 3, 2021.
- Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *International Conference on Learning Representations (ICLR)*, 2024.
- Bo Deng, Yitong Tang, Jiaki Li, Yuxin Huang, Li Wang, Yu Zhang, Yufei Zhan, Hua Lu, and Xiaoshen Zhang. Baseline method of the foundation model challenge for ultrasound image analysis. *arXiv preprint arXiv:2602.01055*, 2026.
- Chiara Di Vece, Zhehua Mao, Netanell Avidris, Brian Dromey, Raffaele Napolitano, Dafna Ben Bashat, Francisco Vasconcelos, Danail Stoyanov, Leo Joskowicz, and Sophia Bano. A multi-centre, multi-device benchmark dataset for landmark-based comprehensive fetal biometry. *Scientific Reports*, 2026. doi: 10.1038/s41598-026-47854-3.
- Alexey Dosovitskiy et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.
- T. Ghi et al. ISUOG practice guidelines: Intrapartum ultrasound. *Ultrasound in Obstetrics & Gynecology*, 52(1):128–139, 2018.
- Jing Jiao, Jin Zhou, Xiaokang Li, Menghua Xia, Yi Huang, Lihong Huang, Na Wang, Xiaofan Zhang, Shichong Zhou, Yuanyuan Wang, and Yi Guo. USFM: A universal ultrasound foundation model generalized to tasks and organs towards label efficient image analysis. *Medical Image Analysis*, 96:103202, 2024. doi: 10.1016/j.media.2024.103202.
- Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 6402–6413, 2017.
- Z. Lin, L. Han, X. Wang, et al. Diagnostic performance of universal-learning ultrasound AI across multiple organs and tasks: The UUSIC25 challenge. *arXiv preprint arXiv:2512.17279*, 2025.
- C. Mitchell, P. S. Rahko, L. A. Blauwet, et al. Guidelines for performing a comprehensive transthoracic echocardiographic examination in adults: Recommendations from the American Society of Echocardiography. *Journal of the American Society of Echocardiography*, 32(1):1–64, 2019. doi: 10.1016/j.echo.2018.06.004.
- Aiden Nibali, Zhen He, Stuart Morgan, and Luke Prendergast. Numerical coordinate regression with convolutional neural networks. *arXiv preprint arXiv:1801.07372*, 2018.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

- Anna Panagiotakopoulou, Alexandros Barmperis, Vasileios E. Katsigiannis, and George K. Matsopoulos. Self-supervised domain adaptation with DINOv2-HRNet for multi-task ultrasound landmark detection. Camera-ready paper, Foundation Model Challenge for Ultrasound Biometry (FU_Biometry), MICCAI 2026 (paper ID MWM_54), 2026.
- Alexandre Sablayrolles, Matthijs Douze, Cordelia Schmid, and Hervé Jégou. Spreading vectors for similarity search. In *International Conference on Learning Representations (ICLR)*, 2019.
- L. J. Salomon et al. ISUOG practice guidelines: Ultrasound assessment of fetal biometry and growth. *Ultrasound in Obstetrics & Gynecology*, 53(6):715–723, 2019.
- Ke Sun, Bin Xiao, Dong Liu, and Jingdong Wang. Deep high-resolution representation learning for human pose estimation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- Gaël Varoquaux and Veronika Cheplygina. Machine learning for medical imaging: Methodological failures and recommendations for the future. *npj Digital Medicine*, 5:48, 2022. doi: 10.1038/s41746-022-00592-y.
- Xiaohua Zhai, Alexander Kolesnikov, Neil Houlsby, and Lucas Beyer. Scaling vision transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12104–12113, 2022.
- Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, and Tao Kong. iBOT: Image BERT pre-training with online tokenizer. In *International Conference on Learning Representations (ICLR)*, 2022.

A. AWS Pricing Calculator entries

The estimate is reproduced at <https://calculator.aws/> by adding the services below in `us-east-1` (On-Demand, Linux, shared tenancy) with the listed *monthly* quantities; the calculator’s monthly total times six months equals the estimate of Table 6 and the requested amount.

Service	Region	Configuration (per month)	USD / month
Amazon EC2	us-east-1	p4de.24xlarge, Linux, On-Demand, 1 instance, 102 h/month	\$2,799.60
Amazon EC2	us-east-1	p5.4xlarge, Linux, On-Demand, 1 instance, 2 h/month (the 12-h pilot)	\$13.76
Amazon EC2	us-east-1	t3.large, Linux, On-Demand, 1 instance, 100 h/month	\$8.32
Amazon EBS	us-east-1	1 gp3 volume, 1,000 GB, 3,000 IOPS, 125 MB/s, 730 h/month, no snapshots	\$80.00
Amazon S3	us-east-1	S3 Standard 1,000 GB-month; 100,000 PUT/LIST, 1,000,000 GET	\$23.90
Data transfer	us-east-1	Outbound to internet 50 GB/month	\$4.50
Monthly total (the calculator’s “monthly cost”); $\times 6$ months = \$17,580.47			\$2,930.08

B. Provenance of the numbers

All figures and tables are generated by `docs/GRNET_AWS/budget.py` from three inputs in `docs/GRNET_AWS/data/`:

- `measured_timings.json`, parsed from the time-stamped epoch lines of our training logs by `tools/extract_timings.py` (64.6 and 15.5 images/s for SSL at 224 and 518 px, 22 fine-tuning runs, 14 probes);
- `bench_results.jsonl`, the ViT-B/L/g micro-benchmarks of `tools/bench_scaling.py`, which replicates the production training step;
- `aws_prices.json`, fetched from the public AWS price list and the ECB reference rate by `tools/fetch_aws_prices.py`.

Re-running the three scripts before submission refreshes every number in this document.